

Geometry Dependence of the Receding Angle of a Droplet on a Solid Cylindrical Surface

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Cite This: *Langmuir* 2025, 41, 10676–10684



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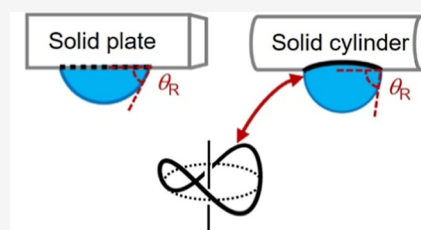
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ABSTRACT: Droplets that partially wet solid surfaces exhibit hysteresis in their contact angle. The values of the minimum (receding) and maximum (advancing) angles, θ_R and θ_A , are empirically well-defined and thought to be unique for a given set of materials. We measured the contact angles of water droplets hanging from hydrophobic, PDMS-functionalized glass and found that the value of θ_R varies with the curvature of the glass. The effect is substantial: θ_R changes from $86.0 \pm 1.9^\circ$ on a flat plate to $95.6 \pm 1.9^\circ$ on a 2 mm diameter rod of the same material. The measured values of θ_A were independent of geometry ($\theta_A = 103.2 \pm 0.9^\circ$). We found a consistent trend among PDMS-functionalized glass cylinders with diameters ranging from 2 to 12.7 mm. We also measured the speed at which the contact line moved just after receding; these results showed a receding speed $\propto \cos(\theta_R) - \cos(\theta_E)$ and a consistent equilibrium contact angle, $\theta_E = 103.4 \pm 2.3^\circ$. Finally, we measured the sliding of water droplets as rods were tilted. The larger θ_R (and thus smaller hysteresis) for a 2 mm-diameter rod led to droplets sliding at a tilt angle of just 21° from horizontal, compared to the 48° minimum tilt for a 7 mm rod. The results show that hysteresis arises from an energy barrier that depends on the shape of the droplet and contact line, both of which change with substrate curvature. The results may lead to designing surfaces that better trap water droplets or shed them for self-cleaning or water-harvesting applications.



Surface shape $\rightarrow$ contact-line shape $\rightarrow$ receding angle

INTRODUCTION

The angle of contact between a solid surface and a fluid interface plays a key role in wetting and is therefore a focus in studies of a wide range of natural phenomena and fluidic technologies. If the shape of the droplet and the contact line can vary freely and if the solid surface is smooth, the contact angle should eventually reach the equilibrium value θ_E , often referred to as the Young–Dupre angle. Crucially, the value of θ_E depends only on the interfacial tensions (γ) of the three interfaces, not on the surface shape.^{1–4} That is, the shape of the solid surface or the presence of body forces such as gravity can change the equilibrium shape of the droplet and of the contact line but not the equilibrium contact angle. Decades of experiments, however, show that the observed contact angles θ in fact do not reach equilibrium in experimentally accessible timescales. Instead, the value depends on the previous motion of the contact line, even in steady state.⁵ Some studies show that contact-angle hysteresis occurs even when the solid surface is clean and no obvious defects and heterogeneity are observed,^{6,7} which suggests that hysteresis is intrinsic to the materials in some cases. A large body of research attributes this hysteresis to a different mechanism: heterogeneity of the solid surface chemistry or topography at length scales that are too small to be observed directly.^{2,8–10} Measuring the contact angle through the liquid phase, experiments report a well-defined minimum contact angle, below which the contact line recedes away from the dry surface. There is also a well-defined maximum angle, above which the contact line advances over

the dry surface. These limiting values are known as the receding (θ_R) and advancing angles (θ_A), respectively. When the contact angle lies in the range between θ_R and θ_A , the contact line remains stationary over a period of minutes to hours. The values of $\theta_{A,R}$ depend on the chemistry of all three phases and the topography of the solid, but are expected to be independent of the droplet shape. Previous studies of contact angle hysteresis have largely focused on droplets resting on or hanging (pendent) from planar surfaces. Indeed, the values of $\theta_{A,R}$ on a flat plate are routinely used to characterize surface coatings, based on the empirical finding that $\theta_{A,R}$ do not depend on the droplet size.^{11–13}

An experimental study has shown that θ_R can, in fact, vary quite strongly with the shape of the water–air interface.¹⁴ In those experiments, a smooth glass sphere was functionalized with polydimethylsiloxane (PDMS), then inserted into air–water interfaces that were planar, hemicylindrical, or saddle-shaped. Using the same sphere repeatedly and changing only the air–water interface shape, those authors found that θ_R was $101 \pm 1^\circ$ at an initially planar interface and was reduced as low

Received: February 24, 2025

Revised: April 9, 2025

Accepted: April 14, 2025

Published: April 21, 2025



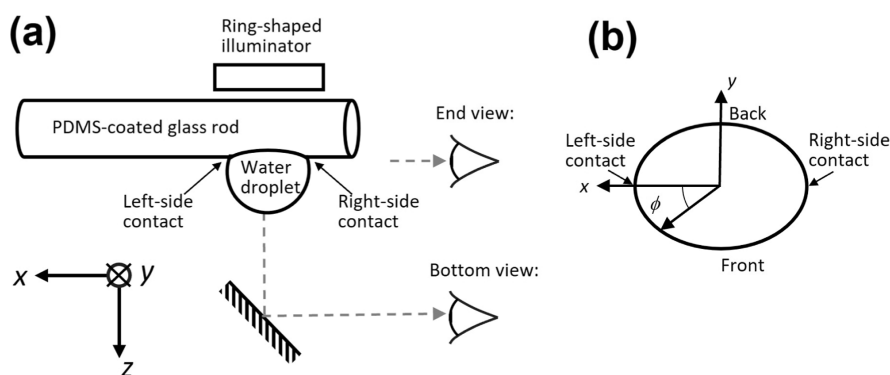


Figure 1. Schematic of the experiment. (a) We used 3 cameras to acquire end-, bottom- and side-view images. The side-view camera imaged the plane shown in this figure (xz plane). The water droplet was placed with a needle from below (not shown). (b) Illustration of the contact line as seen from the bottom. Our measurements report the left- and right-side contact points at $\phi = 0$ and 180° , respectively.

as $80 \pm 1^\circ$ at saddle- or cylinder-shaped interfaces. The value of θ_R varied smoothly with the fluid interface's deviatoric curvature, defined as half the difference between the principal curvatures. Deviatoric curvature, D , is a measure of the anisotropy of the interface shape. As those authors and others have noted, the shape of the contact line depends on D_{fluid} .^{15–19} The change of θ_R , however, was unexpected. A more recent report²⁰ indicates that when a solid sphere sits astride an anisotropically curved interface ($D \neq 0$), the value of θ might vary around the contact line even in the fully receding or advancing case; this behavior would also contradict the standard model of wetting.

We are not aware of any quantitative, predictive explanation for the variation of θ_R with the shape of the interface or contact line. The standard approach to predicting the contact angle considers the virtual work of displacing a small segment of the contact line; such a calculation involves only interfacial tensions and has no mechanism that could lead to a dependence on the shape of the droplet or contact line. A more accurate model of contact-line dynamics, however, considers displacement of a contact-line segment while maintaining mechanical equilibrium everywhere (a constraint involving the Laplace pressure and the interface's mean curvature). Owing to this Laplace pressure condition, displacing any small contact-line segment forces a change of the air–water interface shape, and therefore creates an additional energy barrier against this displacement.^{2,8,21} This mechanism depends on the shape of the contact line and droplet and may account for the observations described above and reported here. We will return to these points near the end of this article.

Here we investigate experimentally how θ_R and θ_A vary with geometry in a different configuration. Instead of inserting a solid sphere into curved fluid interfaces, we place a water droplet on rigid solid surfaces of various curvatures. We use hydrophobic, polydimethylsiloxane- (PDMS-) coated glass in the shape of flat plates and cylindrical rods with diameters ranging from 2 to 12.7 mm. We find that the steady-state advancing angles are similar in all cases, with a mean value of $\theta_A = 103.2 \pm 0.9^\circ$. The receding angles, however, consistently increase with the curvature of the surface: on a planar interface, $\theta_R = 86.0 \pm 1.9^\circ$ and increases to $\theta_R = 95.6 \pm 1.9^\circ$ on a 2 mm-diameter rod. In these pendent-droplet experiments, we also find that the speed at which the contact line recedes (once $\theta = \theta_R$) is slower for the higher-curvature rods. The measured speeds agree quantitatively with the prediction that the force

acting per length of the contact line depends on θ_R . From these measurements, we extract the angle that would correspond to zero speed, which is the equilibrium angle θ_E . Our fit to data yields $\theta_E = 103.4 \pm 2.3^\circ$ for all our substrates, which is indistinguishable from our measured θ_A . This result indicates that the energy barrier for advancing is zero or negligibly small. We then investigated the transition to sliding when the cylindrical rod is tilted away from horizontal. As expected, we find that the droplet slides only when the downhill angle reaches an experimentally well-defined θ_A and the uphill angle reaches θ_R . In agreement with our static measurements, we find that the transition to sliding occurs for a θ_R value that depends on the rod diameter. Because a smaller-diameter rod has larger θ_R , pendent droplets slide with a much smaller rod-tilt angle for a 2 mm rod ($21.3 \pm 0.1^\circ$) than for a 7 mm rod ($48.3 \pm 5.1^\circ$). We attribute the change of θ_R to the shape of the contact line, which becomes increasingly eccentric as the substrate's deviatoric curvature increases. These results are striking and quantitatively significant. We hypothesize that this deformed contact-line shape provides the link between interface geometry and values of θ_R . We describe a qualitative mechanism that may help fill a major gap in our conceptual understanding of contact-angle hysteresis.

For technological applications, our results show how the shape of smooth solid surfaces might be tuned to aid or suppress droplet sliding, which might pave the way to better self-cleaning surfaces and more efficient water harvesting by condensation.^{22–24}

EXPERIMENTAL SECTION

Materials. We used borosilicate glass rods and borosilicate plates purchased from McMaster-Carr. The glass rods with diameters of 2 mm (cat. no. 8496K1), 3 mm (cat. no. 8496K11), 7 mm (cat. no. 8496K15), and 12.7 mm (cat. no. 8496K51) were manufactured by Stemmerich, Inc. The glass plates of 1/8 and 3/16 in. (3.2 and 4.8 mm) thick (cat. no. 8476K35) were manufactured by Abrisa Industrial Glass Inc. We also used borosilicate glass spheres (1.5 and 12.7 mm in radius) with glass rod holders, made by the Scientific Glassblowing Laboratory at the University of Massachusetts Amherst. The spheres are made of Simax glass (a type of borosilicate glass) rods (cat. no. 271010000–271440000) from Friedrich & Dimmock, Inc. The torch temperature for making the glass spheres is 3000 °F. To make the spheres, the Simax glass rods were melted and rotated and a graphite marble mold was used to make the spheres round.

We followed the glass-cleaning and PDMS-functionalization processes described by Şenbil and Dinsmore¹⁵ and Krumpfer and McCarthy.²⁵ All of the rods, plates, and spheres were first cleaned by

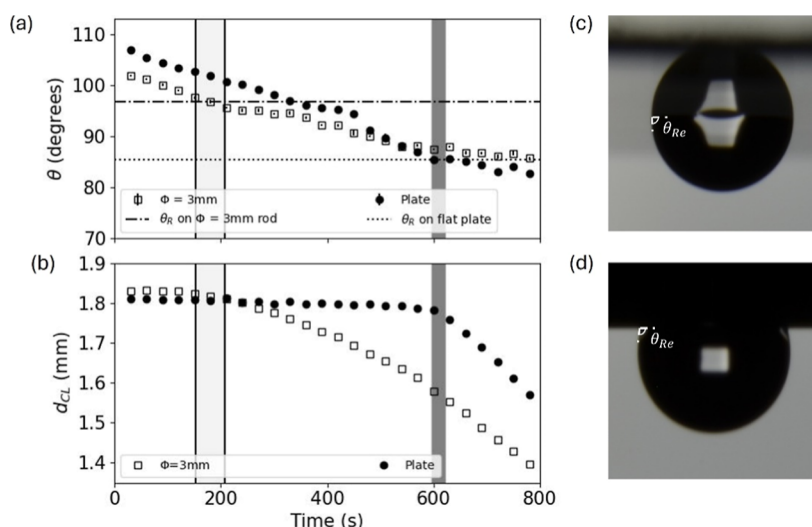


Figure 2. Images and data for receding pendent droplets on a glass plate (●) and a 3 mm-diameter rod (□). When a droplet is placed, evaporation leads to a gradual reduction of θ ; see (a). At first, the contact line does not move and the contact-line diameter, d_{CL} , remains constant; see (b). When θ reaches θ_R , the contact line begins receding. The cross-over from static to receding allows us to define θ_R . Plots (a,b) show θ_R is very different for a flat plate and a rod. (c) A side-view image of a droplet pendent from a flat plate. The glass is near the center and the upper half of the shape is a reflection. (d) A side-view image of a droplet pendent from a 3 mm rod.

soaking in a mixture of Alnochromix (Alconox Inc., cat. no. 2510) and sulfuric acid overnight, then rinsed thoroughly with deionized (DI) water and left in DI water for 2 h. After that, the glass rods, plates, or spheres were blown with nitrogen gas, and dried in an oven at a temperature of 70 °C, then cooled down to room temperature. Next, the glass rods, plates, or spheres were coated by dipping into a 100 mL Pyrex glass bottle filled with PDMS (trimethylsiloxy terminated polydimethylsiloxane, Gelest, Inc., cat no. DMS-T22). After dipping, the glass rods, plates, or spheres were put into empty beakers and left in an oven at 150 °C for 24 h. After heating, they were cooled down to room temperature, washed with toluene, acetone, and deionized water, blown with nitrogen gas, and dried in an oven at 70 °C. The Supporting Information of Şenbil and Dinsmore¹⁵ includes an atomic force microscope (AFM) image of a surface prepared in this manner. They found root-mean-square height variations of 13 nm across a 10 μ m region, with randomly oriented scratches that were a few μ m long and a few nm deep.

Apparatus. As shown in Figure 1a, three cameras were used for this experiment to acquire images from the side, the end, and the bottom of the rod (Nikon model D5100 with DX 18–55 mm lens and a 56 mm extension tube). We used a 30 gauge $\times$ 0.5" blunt tip dispensing needle (CML Supply, item no. 901-30-050) to place a droplet of water. We used a programmable syringe pump (Harvard Apparatus, PHD 4400) to set an initial volume V_{init} at the bottom of the glass plate or glass rod. We then removed the needle. Contact angles were measured from the side-view images. End-view and bottom-view images were obtained to verify that the droplet was centered at the bottom of the rod and to discern when the contact line advances or recedes. The end-view images, however, were not sufficiently clear to provide reliable measurements of contact angles. The bottom-view images (Figure 1b) were used to verify that contact lines were smooth, i.e., without resolvable defects (Figure S1 shows examples.). In most images, each pixel corresponded to 4 μ m in the object plane. The resolution of the camera was estimated at 45 μ m based on images of test objects that had sharply defined right-angle corners. For side-view images, we illuminated from the back side using an LED light panel that was 6.5 cm wide by 2.5 cm tall. For bottom-view images, we found the best image contrast by placing a ring-shaped LED above the sample (1.5 cm diameter by 1 mm wide). In most experiments, we measured the relative humidity using a DHT22 temperature–humidity sensor (Adafruit, Brooklyn, NY) read out by an Arduino. To explore the role of evaporation rate with some of our

measurements, we shrouded the sample in a plastic sheet and used pans of warm water to raise the humidity as high as 72%.

Image Analysis; Measuring θ . The edges of the droplets and the edges of glass rods or edges of glass spheres were extracted from the side-view images using Canny edge detection,²⁶ which was applied through the Python “Canny” function in the “feature” module from the image processing library “Scikit-Image.”²⁷ The parameters sigma, low_threshold, and high_threshold of the Canny function were adjusted until the algorithm found a sharp edge. We verified that the edge defined by the Canny algorithm overlapped the features in the original image. We used the angle tool in ImageJ²⁸ to measure the angles. We established the tangent to the water–air interface using a 100 μ m long line segment constructed from the Canny-edge images. For the pendent droplet on a glass rod, the surface of the rod was clear from the Canny-edge image. For a droplet pendent from a plate, a horizontal straight line was superimposed at the boundary of the droplet and its reflection, which is the bottom plane of the glass plate. For a droplet pendent from a glass sphere, we drew a circle on the image using Adobe Illustrator and constructed a ray tangent to the sphere’s surface. To avoid systematic differences between the left and right sides, all left-side contact angles were obtained from images that were flipped about the vertical direction. Angles were measured separately 3 times for each contact point. We reported the mean and we used the standard error as the measurement uncertainty. As a further test of measurement accuracy, we used the DropSnake plugin to identify the meniscus and extract the contact angle.²⁹ The DropSnake results were indistinguishable from those of the angle-tool method. We found, however, that the DropSnake method is more sensitive to the estimated location of the contact point and hence less reliable. All of the data reported here are from the angle-tool method.

For both advancing and receding image sets, the data were only used if (i) no defects in the contact-line shape were visible and (ii) if the left- and right-side points began receding within 1 min of one another. Most droplets in our experiments (and all of the ones used for the reported data) met these criteria.

Our method amounts to extrapolating the shape at the 100 μ m length scale to obtain a contact angle. This length is much greater than the size of the region that is significantly affected by flow stress, as we will describe in the text below. This length is also much smaller than the two macroscopic scales, which are the droplet radius (1–2 mm) and the capillary length l_c . (l_c is defined as $\sqrt{\gamma/\Delta\rho g}$, where γ is the liquid–vapor interfacial tension, g is the acceleration due to gravity, and $\Delta\rho$ is the difference in mass densities of water and air. Its

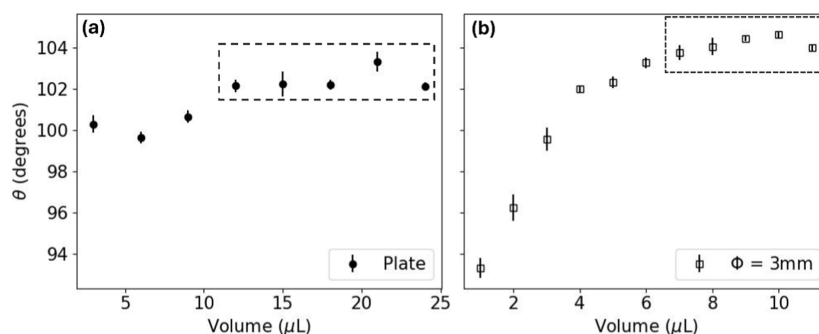


Figure 3. Contact angle θ during the advancing process as the volume of pendent droplet was gradually increased. The advancing contact angle θ_A for pendent droplets are calculated as the mean of the data points inside the dashed boxes. (a) For the flat plate, $\theta_A = 102.4 \pm 0.4^\circ$. (b) For the 3 mm-diameter rod, $\theta_A = 104.2 \pm 0.3^\circ$.

value is approximately 2.7 mm). In all cases, the observed interface shapes are smooth and convex as expected.

RESULTS AND DISCUSSION

Figure 2 shows examples of a pendent droplet receding on a flat plate and on a 3 mm-diameter rod. In most cases, the droplet's initial volume, V_{init} , was 5 μL . There are 4 trials where V_{init} was 10 or 15 μL and we found that the resulting measured θ_R did not depend on V_{init} .

After placing each droplet, we observed as it slowly shrank due to evaporation. A set of side-, end-, and bottom-view images was captured every 30 s. From the side view (xz plane), we observed that θ at first decreased over time (Figure 2a). During this stage, the distance between left and right contact points, d_{CL} , remained constant and each contact point remained static; the droplets were not sliding. After a time on the order of 100 s, we observed the left and right contact points move in the receding direction. At the same time, θ reached a plateau value or transitioned to a slower decay rate. Figure 2 shows that a transition from static to receding contact occurred on the flat plate near $t = 600$ s at an angle defined as θ_R (here, $85.4 \pm 0.4^\circ$). On the 3 mm rod, the transition occurred near $t = 200$ s at an angle $\theta_R = 96.8 \pm 0.3^\circ$. The difference in receding angles between these two geometries is clear from this plot. We repeated this experiment several times, with results described below.

From the end-view images, we established the time at which the contact line moved at the front and back positions, defined by $\phi = 90$ and 270° in Figure 1b. For all of the pendent-droplet data described below, we found that the contact line began receding at approximately the same time at all four symmetry points, $\phi = 0, 90, 180$, and 270° .

According to the standard picture of contact-angle hysteresis, a receding experiment like ours should find two stages of behavior: d_{CL} should remain fixed as θ decreases until θ reaches θ_R . Thenceforth, d_{CL} decreases and θ remains constant. Prior studies of droplets evaporating on a flat plate showed that indeed the transition from static to receding contact is well defined but θ was still gradually decreasing in the receding stage, even with pure water.³⁰ Those authors found that adding contaminants to the water caused θ to change more rapidly in the receding stage. This observation suggests, but does not prove in general, that the gradual decrease arises from accumulation of contaminants at the contact line even with nominally pure water. We investigated the possibility of contaminant deposition by adding micron-sized polystyrene tracer beads to the water droplet. On both a

flat plate and a rod, we observed the tracer beads moving in a random fashion (presumably driven by room air currents), with no preferential flow toward the contact line. This finding differs from previous coffee-stain studies³¹ because the surface was clean enough that the contact line could move smoothly.

To probe the advancing process, we injected water through the needle into a droplet at the bottom of the glass plate or glass rod. We injected a programmed volume ΔV in each step. ($\Delta V = 3 \mu\text{L}$ for plates, $\Delta V = 2 \mu\text{L}$ for 12.7 mm-diameter rods, $\Delta V = 1 \mu\text{L}$ for 7 and 3 mm rods, and $\Delta V = 0.5 \mu\text{L}$ for 2 mm rods.) After adding water, we removed the needle and waited 90 s to let the shape of the droplet become stable, and then obtained images of the side view, end view, and bottom view. The 90 s relaxation time was long enough so that flow perturbations from removing the needle were no longer visible. (In control tests, we found that the evaporation during this 90 s either had no effect on the measured θ or reduced it by not more than 2° .) The cycle of inject, wait, image was repeated. At first, the measured values of θ increased with each cycle. The distance between the two visible contact points, d_{CL} , also increased for each step even when θ was relatively low, which we attribute to vibrations of the needle causing water flow. After a few steps, the measured θ increased to a plateau value. We repeated the water addition cycles and obtained images at least 5 times after θ reached the plateau. We defined θ_A as the average of these five values (see the dashed boxes of Figure 3). Example results for a flat plate and a 3 mm-diameter rod are shown in Figure 3.

Figure 4 shows the measured advancing and receding angles of pendent water droplets on PDMS-coated glass in the shape of flat plates and cylindrical rods. Each open symbol shows the result of one droplet, the average of the left and right sides. The filled points show the mean value for each glass shape. In all cases, we verified that the contact line was smooth at a scale of 50 μm and larger. We repeated measurements using the same glass rod after repeating the cleaning and PDMS coating process; we found no systematic variation of angles. For receding measurements at $D = 0.33 \text{ mm}^{-1}$ (3 mm diameter), the top 4 data points are from one rod and the lowest 2 data points are from a different rod. The lowest data point also had the greatest left-right difference among our data; this difference suggests that this data point might be affected by a heterogeneity on the glass but there was no direct evidence to justify excluding this data. Overall, we find that the advancing angle is consistent for all of our shapes, with a mean value of $103.2 \pm 0.9^\circ$.

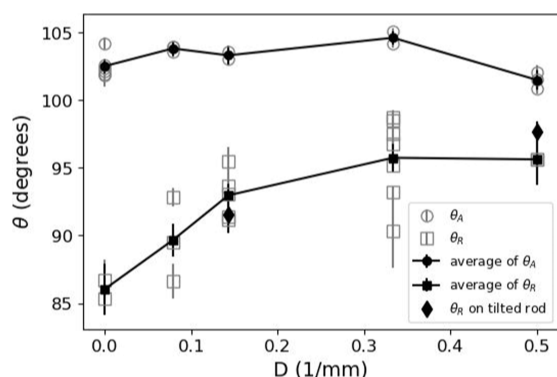


Figure 4. Measured advancing and receding contact angles of pendent water droplets on horizontal PDMS-coated glass surfaces. The horizontal axis is the deviatoric curvature, D , of the interface, equal to $(\text{rod diameter})^{-1}$. A flat plate was used for $D = 0$. The values of θ_A (upper data) are constant. The values of θ_R (lower data) increase significantly with D . Open circles and squares are the results of individual droplets. The smaller filled symbols, connected by line segments for visibility, show the mean value for each D . The diamonds for $D = 0.14$ and 0.5 mm^{-1} were extracted from the tilted-rod data of Figure 7.

By contrast, the receding angles vary systematically with the shape of the glass surface (Figure 4). Following the approach of Şenbil and Dinsmore, we quantified the glass shape by means of its deviatoric curvature, D , which is defined as half the difference between the principal curvatures at the point where the droplet resides. For plates or spheres, $D = 0$, while for rods $D = 1/d$, where d is the rod diameter. In essence, D captures anisotropy of the curvature. Şenbil and Dinsmore pointed out that D sets the contact-line shape (as it does here; see below), which is likely the mechanism by which θ is affected. Here we find that the receding angle increases significantly with D .

The mean curvature (half the sum of the principal curvatures) also varies with cylinder diameter, so Figure 4 by itself does not distinguish which parameter is relevant. To investigate whether mean curvature (H) might play an important role, we carried out control experiments with droplets hanging from glass spheres, which have $D = 0$ and different H . As described in the Experimental Section, these spheres were made of borosilicate glass from our glass shop because a suitable range of sizes was not available from the supplier of our glass rods and plates. Following the same surface preparation and measurement procedures, we measured θ_R for two spheres that have $D = 0$ and $H = 0.079$ and 0.67 mm^{-1} (and dimensionless $d_{\text{CL}}H$ in the range of 0.15 to 0.91; Figure S2). We measured average θ_R values of $90.9 \pm 0.4^\circ$ and $90.0 \pm 1.5^\circ$, respectively. These values are indistinguishable, so we conclude that θ_R did not depend on sphere mean curvature, H . (We note, however, that these values differ slightly from our glass plates, which we attribute to the use of a different glass even though the PDMS coating was applied by the same procedure.) We therefore conclude that D , not H , is the key parameter that influences θ_R . Şenbil and Dinsmore reached the same conclusion for the case of solid spheres at curved air–water interfaces, supported by their measurement of θ_R on saddle and cylindrical shapes that had similar D but different H .

The primary result of this article is that the receding angle increases significantly with D , while the advancing angle does

not. In particular, we found that θ_R increased from $86.0 \pm 1.9^\circ$ on a flat plate to $95.6 \pm 1.9^\circ$ on a 2 mm-diameter rod ($D = 0.5 \text{ mm}^{-1}$). The advancing angle in all cases remained steady, with an average value $\theta_A = 103.2 \pm 0.9^\circ$.

Receding Speed on Horizontal Rods. In our receding experiments, we measured the speed, v_{CL} , of the contact line just after the crossover from pinned to receding. We defined v_{CL} as the time derivative of d_{CL} , the distance between the left- and right-side contact points. We obtained v_{CL} by a linear fit to the first five measurements of d_{CL} after the contact line began receding. The data in Figure 5 show a pronounced trend, in

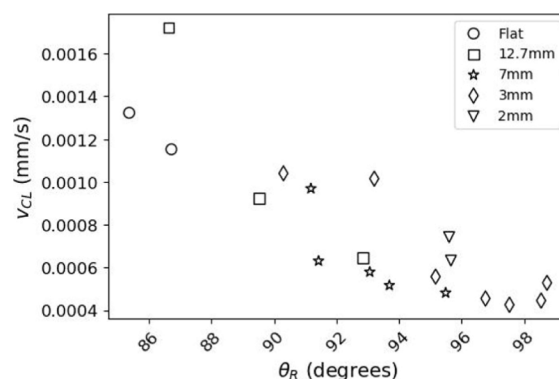


Figure 5. Measured receding speed, v_{CL} , of sessile droplets just after the transition to receding. Each symbol represents the result for one droplet. Droplets were pendent from flat plates and rods whose diameters are shown in the plot legend. Extrapolating to $v_{\text{CL}} = 0$ yields an estimate of $\theta_E = 103.4 \pm 2.3^\circ$.

which larger values of θ_R resulted in slower v_{CL} . The change is substantial: increasing θ_R by 13° resulted in a roughly 4-fold decrease of v_{CL} . The relative humidity of these measurements varied from 16% to 72%. We found that relative humidity had no systematic effect on the measured θ_R or on v_{CL} . We can explain our results using a simple equation of motion for the contact line. Following the classical argument of Young, the force per unit length of the contact line is $\gamma(\cos(\theta) - \cos(\theta_E))$.² In the overdamped limit, this capillary force is equal and opposite to the viscous friction, which is proportional to ηv_{CL} where η is the shear viscosity of water. Looking at the contact-line speed at the receding angle, we therefore expect a linear relation between v_{CL} and $\cos(\theta_R)$. Figure S3 shows that our data follows this linear trend, which suggests that all of our substrates have the same θ_E (as expected). Furthermore, we extrapolate a linear fit of our data to $v_{\text{CL}} = 0$, which corresponds to $\theta = \theta_E$. From this analysis, we estimate that $\theta_E = 103.4 \pm 2.3^\circ$. This value is indistinguishable from the average measured θ_A ($=103.2 \pm 0.9^\circ$). The similar values of θ_A and θ_E imply that the energy barrier (or force barrier) against advancing is negligibly small. This finding explains why θ_A has no discernible dependence on geometry; we return to this point below.

Contact Line Shape. Following the logic of Şenbil and Dinsmore,¹⁵ we propose that θ_R is sensitive to geometry because of the shape of the contact line itself. (We return to this point in the final section.) It is well-known that droplets partially wetting surfaces of nonaxisymmetric shapes form nonspherical shapes and noncircular contact lines. The shape of the contact line has been predicted for a solid sphere on an initially cylindrical fluid interface.^{32,33} To leading order in $d_{\text{CL}}D$, the contact line height has a quadrupolar shape,

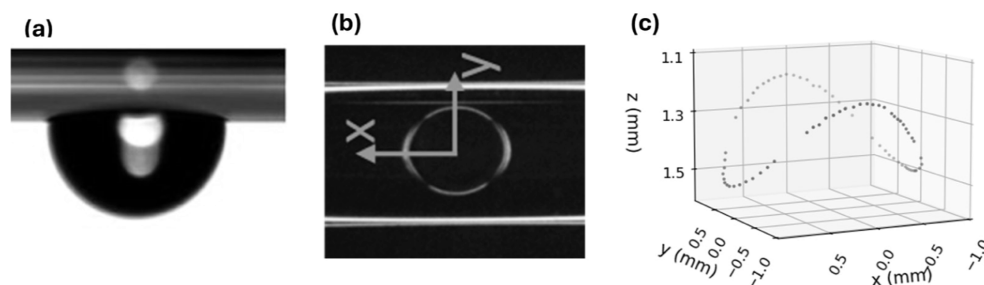


Figure 6. Images of a pendent droplet on a 3 mm-diameter rod. (a) Side view, with illumination that shows the contact line. (b) Bottom-view image, showing the ellipse-shaped contact line. (c) The 3d plot of the contact-line shape obtained by projecting the bottom-view shape onto the cylinder. The contact line is elongated along the rod (x -axis) and the height has the characteristic quadrupolar symmetry.

$z_c = \frac{1}{24}D_0d_c^2 \cos(2\phi)$, where d_c is the diameter of the contact ring and D_0 is the deviatoric curvature of the interface before the sphere adsorbs.³² Here we find that the “reverse” case of a fluid droplet on a solid cylinder also induces an undulating contact line. Figure 6 shows images of a receding droplet pendent from a 3 mm-diameter rod. By projecting the imaged contact line onto the surface of the cylinder, we obtain the full 3D shape of the contact line (Figure 6c). The variation of contact-line height (z) has a clear quadrupolar component proportional to $\cos(2\phi)$, elongation along the x -axis, and perhaps other modes as well. Figure S4 shows an example of a droplet on a 12.7 mm-diameter rod, which had a similar contact-line shape.

Droplets Sliding along a Tilted Rod. Motivated by our static data of Figure 4, we investigated the sliding of droplets along a tilted rod. The standard interpretation of a droplet on a surface that is increasingly tilted is that the uphill angle decreases and the downhill angle increases. When both angles reach their respective limits, then the entire contact line moves and the droplet slides downhill. (Previous work has shown that the uphill and downhill limiting angles are not necessarily equal to θ_R and θ_A ¹⁰ but should be so at the sliding threshold.³⁴) Our findings of Figure 4 motivated us to investigate whether a droplet would slide more readily on the higher-curvature rod because of the smaller difference between θ_R and θ_A .

We obtained images of 5 μ L droplets pendent from 2 to 7 mm rods as the rod tilt angle was varied. We first set the rod tilt angle (β , defined relative to horizontal). We placed a 5 μ L droplet, then imaged over a period of 5 min to see if it slides. Finally, the droplet was removed by wicking into a clean tissue, the rod was rotated along its axis (maintaining β), and a fresh droplet was placed. This process was repeated for at least 3 droplets. Figure 7 shows the uphill angles (black) and downhill angles (gray) vs rod tilt angle β . In each case, the open symbols show droplets that were fixed in place. Filled symbols show droplets that were sliding (meaning that both uphill and downhill contact points were moving).

As expected, we found that the downhill angle increased with β and the uphill decreased, until the point where the droplet slid. As shown in Figure 7, the 2 mm rod droplets showed a transition from static to sliding when the uphill angle fell below approximately 98° (dashed line). By contrast, droplets on the 7 mm showed this transition at approximately 92° . We defined $\theta_{A,R}$ as the measured downhill and uphill angles at the static-to-sliding transition point. The receding angles extracted from this analysis are $91.5 \pm 1.3^\circ$ and $97.7 \pm 0.8^\circ$ for the 7 and 2 mm rod, respectively. These values are

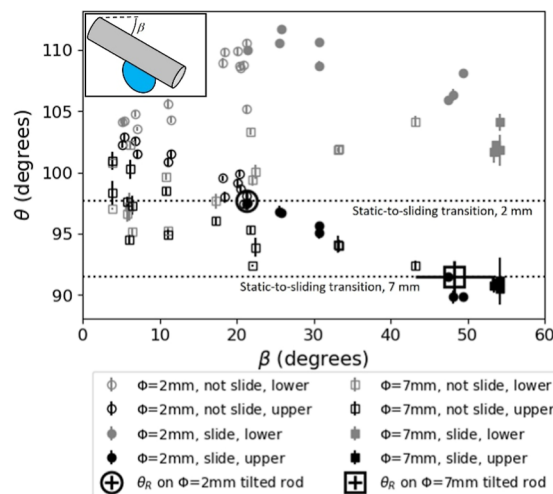


Figure 7. Droplets pendent from a rod tilted by an angle β from horizontal (illustrated in inset). Each symbol represents the contact angle on the most uphill point (black) or downhill point (gray). Open symbols are static, while filled symbols show sliding droplets. The dashed lines show the transitions from static to sliding. The large circle and square highlight the (receding) angles, measured at the static-to-sliding transition. The result for the 2 mm rod is greater than for the 7 mm rod.

shown as the large symbols in Figure 7 and as the diamonds in Figure 4. With this tilted-rod geometry, we find again that θ_R increased with rod curvature, quantitatively consistent with our nearly static, horizontal-rod data.

We also found that droplets on the small rod slide at a much smaller tilt angle than on the large rod: $\beta_{\text{slide}} = 21.3 \pm 0.1^\circ$ and $48.3 \pm 5.1^\circ$ for the 2 and 7 mm rods, respectively. This result is explained by the lower hysteresis of the smaller rod. As we discuss below, this finding has important implications for droplet sliding on filaments and threads.

The downhill angle remains approximately constant near 105° for the 7 mm rod, which is reasonably close to the static θ_A for the horizontal rod. For the 2 mm rod, however, we repeatedly found that the downhill angle rose as high as 112° without sliding. We argue that the downhill contact line remained static because the uphill angle was still too high to recede; therefore the droplet further distorted toward the downhill direction, increasing the downhill angle. This finding highlights the known—but still sometimes overlooked—point that advancing and receding processes are not determined by a local condition that applies at every point. Rather, the transition to advancing or receding involves the entire contact line.²¹ We return to this point in the next section.

In general, the shape of the droplet and the apparent contact angle could differ for a static droplet compared to an evaporating or sliding droplet. In all cases, our system is in the viscous-dominated limit, as shown by the small value of the Reynolds number, $Re \equiv d_{CL}\nu\rho/\eta \approx 10^{-3}$. In this expression, ρ and η are the mass density and shear viscosity of water, and ν is taken to refer to ν_{CL} , the contact-line speed just after the receding threshold, from Figure 5. The Capillary number of the receding droplets compares the stresses from viscous vs capillary effects: $Ca \equiv \eta\nu_{CL}/\gamma$. Typical values are $Ca \approx 10^{-8}$ in all of our experiments. Even though $Ca \ll 1$, there is expected to be a region near the contact line where the interface shape is perturbed by flow-induced stress. The size of this region is roughly set by $d_{CL}Ca/\theta$, which is far below the resolution of our camera and thus not visible in our experiments.^{2,35} Therefore, at the scale of all of our measurements, the droplet shape and apparent θ are negligibly affected by the fluid flow. Earlier experimental studies also found that the capillary forces and angles were independent^{36,37} of or only weakly dependent³⁸ on the contact line speed in the same range of ν_{CL} as here. The measured $\theta_{A,R}$ should therefore be equivalent to the ideal static, stationary case.

CONCLUSIONS

We showed that the anisotropic (deviatoric) curvature of a hydrophobic solid surface can change the receding contact angle θ_R but not the advancing angle θ_A . In pendent-drop experiments on horizontal rods, the contact line recedes or advances everywhere. By contrast, the sliding droplets on tilted surfaces recede on the uphill side and advance on the downhill side. In all of these cases, the steady-state receding angles changed with surface shape. We propose that the key is the smooth undulation of the contact line caused by the anisotropy of the interface. As the solid substrate changes from a flat plate ($D = 0$) to a cylindrical rod ($D > 0$), the contact line changes from a circle to (approximately) the rim of a saddle. The effect is significant: the steady-state value of θ_R increased by nearly 10° as D increased from zero to 0.5 mm^{-1} .

Extracting the equilibrium angle from our receding-speed data, we found that the plate and the rods have a common value of $\theta_E = 103.4 \pm 2.3^\circ$. This value is indistinguishable from the average θ_A ($103.2 \pm 0.9^\circ$), which suggests that the advancing process has a low or no barrier or dissipated energy, whereas the receding process has a large (and geometry-dependent) energy barrier. Prior work (such as refs 36 and 39) attributed the advancing-process energy barrier to nonwetting, hydrophobic spots on the surface; our result implies that these defects are rare, small, or energetically weak. The receding-process energy barrier may arise from defects that are wetting or hydrophilic,^{36,39} perhaps including nanoscopic regions of exposed glass. Future scanning-probe or other nanoscopic surface characterization to investigate this point would be useful.

In our experiments, the droplet size and the range of rod diameters were all comparable to the capillary length, l_c . Gravity and other body forces change the droplet shape but not θ_E . We expect that the contact-line shape and therefore $\theta_{A,R}$ will vary with $d_{CL}D$ even at scales much smaller than l_c (e.g., based on refs 32 and 33). The same trend may apply even to wetting of nanoscopic fibers,³⁷ though we note that on such small structures, the small value of d_{CL} might offset the large value of D , so that $d_{CL}D$ may be in the same range as

reported here. This would be an interesting topic for future experiments.

The typical approach to predicting the contact angle considers a short segment of the contact line and ignores the rest of the droplet. If $\theta \neq \theta_E$, then there is a net force on this segment, acting tangent to the solid surface with magnitude $\gamma[\cos(\theta) - \cos(\theta_E)]ds$, where ds is the segment length.^{2,36,40} A simple way to account for contact-angle hysteresis is to assume an irreversible energy cost per area, Γ_A or Γ_R , required to re- or dehydrate a unit area of solid surface and allow the line to advance or recede.² The surface might be heterogeneous but we assume that Γ has a well-defined average over a region that is large compared to molecular size and small compared to the droplet size d_{CL} and the capillary length l_c . The work to displace this line a distance dl along the solid surface may be compared to the irreversible energy cost, ΓdA , where dA is the area swept out by the moving contact line segment ($=dsdl$). Defining θ_R as the angle where the work equals the energy lost, so that the line can move without energy cost, we find $\cos(\theta_R) = \cos(\theta_E) + \Gamma_R/\gamma$. Therefore, $\theta_R \leq \theta_E$, consistent with observations. Similarly, for advancing, we find $\cos(\theta_A) = \cos(\theta_E) - \Gamma_A/\gamma$ and hence $\theta_A \geq \theta_E$. This model (or any other that considers a segment of the contact line in isolation) incorrectly predicts that contact angle hysteresis is independent of the contact line geometry.

More likely, the steady-state contact angle should be set by a *nonlocal* condition owing to the strong constraint on the mean curvature H of the fluid interface. That is, if only a segment of the contact line moves at a given moment, then the shape of the fluid interface must change. The spatial extent of this change could be large, likely set by the smallest of l_c , d_{CL} , or the curvature of the displaced segment of the contact line. The energy cost of this interface deformation contributes to the energy barrier needed to move the contact line.^{2,8,21} This point has been made previously, though it is still frequently overlooked. With this model, if we now change the global shape of the contact line and again move a small segment of the contact line, then the shape of surface deformation changes too. We should therefore expect that the energy barrier to moving this line segment also changes, depending on the initial shape. Our experiments show that this energy barrier decreases when the contact line deviates from circular, so that the measured θ_R approaches θ_E . Nonetheless, the *quantitative* connection between contact-line shape and energy barrier for advancing and receding has yet to be worked out. For PDMS-coated glass, we now know of one case where curvature anisotropy increases hysteresis (glass spheres in deviatoric interface¹⁴) and one where it decreases hysteresis (droplets on deviatoric solids, this paper). Together, these results provide an excellent testing ground for theoretical models, an important area for future research.

We found compelling data that geometry affects advancing and receding angles and droplet motion for PDMS-coated glass. We do not know if this trend applies to other surfaces as well, and we hope that future investigations will answer this question.

Our findings have important practical or technological implications. When a colloidal particle adsorbs at a fluid interface, the steady state contact angle might depend on the shape of the particle. Droplet condensation collectors might work better with smaller-diameter filaments because the threshold droplet size for sliding will be smaller on the thinner filaments. Alternatively, it might be that a different choice of

filament cross-section shape (but the same surface chemistry) would give further control of droplet sliding. Likewise, tilted slabs that have smoothly patterned features may become more effective at shedding water droplets, leading to better self-cleaning surfaces.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.5c00936>.

(I) Examples of contact lines with defects, which are not used for analysis. (II) Method of extracting the receding angle and uncertainty from the data. (III) Receding angles of a droplet pendent from a PDMS-coated glass sphere: varying H with constant D . (IV) Receding speed on a horizontal rod vs D . (V) The contact-line shape inferred from bottom-view images (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was funded in part by a grant from the National Science Foundation (CBET-1803797). We thank Sally Prasch of the UMass Amherst Scientific Glassblowing Laboratory for making the glass spheres, and Chris Oville, Nuoya Zhou, Narayanan Menon, Benny Davidovitch for helpful discussions.

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